



Society of Petroleum Engineers

SPE-229816-MS

Estimation of Regional Heat Flow Through a Data-Driven Methodology: From Basin Modelling to Rare Earth Elements Exploration

Abul Khair M Fahimuddin, Walid Hussein, and Waleed Saleh, Saudi Aramco

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229816-MS](https://doi.org/10.2118/229816-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Estimating regional heat flow is of paramount importance for oil and gas exploration, as it directly influences the thermal maturity of source rocks, and impacts reservoir quality predictions. Additionally, heat flow plays a crucial role in exploring rare earth minerals (REMs) from geothermal brines by identifying areas with high subsurface temperatures and active fluid circulation. Hence numerical estimation of regional heat flow distribution is a key element for subsurface workflows within exploration. Subsurface heat flow (q) is typically calculated using Fourier's law of heat conduction for a layered Earth model: $q = \sum_i \lambda_i \nabla T_i$ where λ_i is the thermal conductivity of the i -th layer, and ∇T_i is the temperature gradient across that layer. There are several limitations of such methods such as non-incorporation of complex geology and lack of available temperature. A dual-mode heat flux estimation is proposed here by combining physical modeling (Fourier's Law) with predictions from a trained Random Forest ML-model. Using synthetic regional data comprising lithology, surface temperature, geospatial location and Horner-corrected bottom hole temperature data, the predicted regional heatflux demonstrating promising results. The proposed methodology has been implemented as a representative synthetic use case for estimating the regional heat flux where there are several representative wells with temperature data available at various depths alongside average annual land surface temperature reading at well sites. A depth-dependent (lithology variation) thermal conductivity is applied to calculate thermal resistance, which together with corrected temperatures and land surface temperature, allows for the estimation of heat flow via Fourier's Law. To implement data-driven generalization to predict heat flux at unseen locations, both spatial (latitude/longitude) and physical (depth, surface temp, temperature) data were used as ML-features to train through Random Forest regressor. Accuracy of ML training was ensured through root mean squared error (RMSE) and mean absolute error (MAE) calculation. To validate the ML forecasted results, a quantitative comparison of predicted heat flux was utilized with respect to physical law-derived heat flux values. To enhance spatial realism and prevent model overfitting, RF prediction is blended with the spatially interpolated inverse distance weighting estimate, and further augmented through application of radial basis functions (RBF). The temperature-depth profile of the predicted well aligns with the established regional thermal trend observed across the training wells. The estimated heat flow also falls within the upper range of the training dataset, reinforcing the notion that this predicted location lies within a regional high heat flow corridor. Thus, the proposed methodology

proved predictive capability of combined data-driven and physics-based modeling to estimate regional heat flow. It has the potential to support more robust decision-making in oil and gas basin modeling, thermal regime studies and for exploring rare earth minerals (REMs) from geothermal brines, especially in under-characterized areas or frontier regions with limited temperature data.

Introduction

Accurate understanding of the thermal regime in sedimentary basins, is of fundamental importance to reconstructing the thermal maturation history of source rocks, thus making heat flow prediction a vital task within the energy industry (Palumbo 1999). The maturity of source rocks depends on their time-temperature exposure, while heat flow directly shapes the geothermal gradient that governs this process (Qiu 2022). Higher heat flow accelerates mineral diagenesis, in turn, alters porosity and permeability over geological processes, thereby playing an important role in determining reservoir quality. Similarly in mineral exploration, especially for rare earth minerals (REMs) sourced from geothermal brines, accurate heat flow modeling plays a pivotal role in identifying and evaluating prospective exploration zones (Smith 2017). Usually, high heat flow regions correspond to deep, and high-temperature reservoirs where circulating reservoir fluids can dissolve and transport REMs (Szanyl 2023). In addition, the interaction between heat flow and specific lithologies exerts fluid–rock reactions which dictates leaching and subsequent mineral deposition (Dobson 2023). It is worth to mention that basin-scale regional heat flow modeling also facilitates long-term brine productivity prediction which could be done by assessing recharge rates of these mineral-rich fluids. Furthermore, elevated heat flow is frequently associated with geological structures such as faults or crustal thinning zones, indicating conduits for deep hydrothermal circulation, potential pathways for metal mobilization and concentration (Cetin 2024). Despite their foundational value, these traditional approaches face limitations in predicting heat flux, and the reasons are attributed to sparse data coverage, simplified assumptions about thermal properties, and weak integration with multidimensional geological information (Smith 2017). This highlights the necessity for more advanced, and preferably, data-driven techniques for determining regional heat flux distribution within modern energy and mineral exploration workflows.

The foundation of heat flow calculation within the subsurface is based upon Fourier's law of heat conduction, applied in a layered model where thermal conductivity varies with depth due to different rock types. With the assumption of steady-state heat conduction (i.e., no transient effects) and each earth layer having constant thermal conductivity based on rock types (shale, carbonates, sandstone, or crystalline basement), the heat flux vector $\mathbf{q}=(q_x, q_y, q_z)$ can be formulated as:

$$\mathbf{q} = -\lambda \nabla T$$

Where $\lambda(\mathbf{x}, z)$ is the thermal conductivity at position $\mathbf{x}=(x, y)$ and vertical depth z , and $\nabla T = \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right)$ is temperature gradient. Thus, the vertical heat flow at a point $(\mathbf{x}, z)$ can be written as

$$q(\mathbf{x}, z) = -\lambda(\mathbf{x}, z) \frac{\partial T}{\partial z} \quad (1)$$

In practice, subsurface heat flux is calculated using various data types, each providing inputs for estimating the temperature gradient and thermal conductivity. Temperature data, typically obtained from borehole measurements such as bottom-hole temperatures (BHTs) or continuous temperature logs, are used to determine the temperature gradient. However, since BHTs are affected by drilling disturbances, they require correction to reflect true formation temperatures (Fahimuddin 2024). Thermal conductivity which is a key component described in equation (1), can be sourced from direct laboratory measurements on core samples. As a common practice, it is inferred from rock type or petrophysical logs. Different rock types, such as shale, sandstone, or carbonate, exhibit typical thermal conductivities values, that allow estimation of

thermal conductivities where lithological logs or cuttings are available. Additionally, empirical relationships exist to ascertain conductivity from well logs like sonic, density, or resistivity logs. Stratigraphic and lithologic information also plays an important role when it comes to estimating subsurface heat flow, especially in layered earth models, where heat flow is calculated across multiple geological units with different conductivities. In such cases, a composite heat flow estimates is the optimal way that accounts for both layer thickness and conductivity. Obtaining reliable temperature data with depth, is critical for resolving changes in geothermal gradients, and infer aquifer boundaries. Surface temperature data, usually obtained from climate models or weather stations, serves as the upper boundary condition for heat flow models. In more detailed analyses, internal heat sources, such as radiogenic heat production from uranium, thorium, and potassium in basement rocks, may be included to refine the vertical thermal profile (Qiu 2022). When combined thoughtfully, these disparate inputs allow for robust heat flow estimation that supports geothermal resource assessment, basin modeling, and thermal history reconstruction relevant for rare earth minerals (REMs) from geothermal brines (Dobson 2023). A key approach is to identify areas with high subsurface temperatures and active fluid circulation, thus enabling the location of further exploration activities to be determined.

Recent advances in data-driven techniques, leveraged through machine learning (ML) methodologies, for determining regional heat flux distribution have significantly enhanced predictive accuracy and spatial resolution. Machine learning (ML) is a subfield of AI, which uses statistical methods to enable "machines" to improve with experience (in this context, "machines" are computer programs and algorithms). These programs or algorithms are designed in such a way that they can learn and improve over time when exposed to new data. ML technologies are used everywhere, from simple spam email identification to more sophisticated generative AI applications, and there is plethora of ML-applications within oil and gas industry (Sircar 2021). Both the geothermal sector and REMs domain have observed considerable interest in ML applications in recent years. For a comprehensive overview of ML applications within geothermal resources, it is recommended to consult (Okoroafor 2022).

The key breakthroughs have been in the realms of integrating heterogeneous datasets such as satellite observations, well logs, and climate models, within ML frameworks to predict heat flux within the subsurface (Shahdi 2021). Recently, researchers successfully applied XGBoost models using geological and geophysical features to predict geothermal heat flux with high accuracy (Zhang 2025). For larger regional scale studies in the northeastern United States, it has been reported that the usage of XGBoost and Deep Neural Networks (DNN) models to predict bottom-hole temperature logs attained very high degrees of accuracy, thus aiding regional geothermal exploration (Aljubran 2024). Software frameworks such as GeoThermalCloud, are combining unsupervised and supervised ML-techniques with physical constraints to unravel new geothermal systems (Mudunuru 2022). Within the rare earth mineral exploration domain, ML-methods such as the emergent self-organizing map (ESOM), have been recently applied for multidimensional interpretation and prediction of rare earth element (REE) abundance in produced and geothermal waters in the United States (Engle 2022).

The interpolative physics-informed graph neural network (InterPIGNN) model proposed by (Aljubran 2024) represents a state-of-the-art approach to directly predicting regional heat flux. It integrates diverse geophysical, geological, and thermal datasets (such as BHT, conductivity, gravity, and tectonic proximity), where these data sets are structured into a graph neural network that respects the underlying steady-state heat conduction physics. By combining data-driven learning with physical constraints, the model produces high-resolution 3D temperature and heat flow fields across the continental U.S. However, without sufficient thermal measurements or surface heat flow labels, the model's predictive performance may be reduced unless it is supplemented with auxiliary geophysical inputs such as gravity, crustal thickness, or radiogenic heat data. Thus, while InterPIGNN can guide the regional understanding even in low-data scenarios, its effectiveness depends on the availability of broader contextual datasets or pre-trained models. In the

context where only basic inputs like well temperature, depth, and lithology are available, simpler hybrid approaches such as Random Forest models, can be used to estimate temperature gradients combined with thermal conductivity to compute heat flow, may be more practical. The proposed methodology in this paper addresses the need to formulate data-driven method for predicting regional heat flow where there is a lack of pervasive datasets as required by methods like InterPIGNN. The first step begins by integrating available diverse subsurface and surface data inputs, including temperature logs (e.g., Horner-corrected bottom-hole temperatures), lithological information, thermal conductivity estimates, and spatial features like coordinates and elevation. Then, these inputs are fed into a heat flow prediction model (Random Forest regressor, in this case). This ML-model, coupled with geospatial interpolator (e.g., kriging or inverse distance weighting), estimates the regional heat flux across unsampled areas, away from given well locations. The predicted heat flow map is then used to identify high heat flow zones, which are, potentially along with other factors such as radiogenic heat, that are potential exploration targets for geothermal brine accumulation. These zones, particularly when aligned with favorable geology and permeability structures, offer ideal conditions for mineral deposition (Cetin 2024). By training on observed heat flux values (e.g., from wells with complete data), ML-algorithms such as Random Forests, Gradient Boosting, or Neural Networks can learn correlations between these multi-source features and the corresponding calculated heat flow. Once trained, these models can extrapolate to predict heat flux at unsampled locations using only the input features available at those sites. Additionally, the proposed ML-based formulation enables uncertainty quantification and feature importance analysis. In the following sections, attempts are made to establish the proposed workflow, followed by a detailed description of the synthetic use case, and finally, explain the outcome of the ML-experiments, and its applicability to regional heat flow prediction.

Methodology

In real-world exploration, we often aim to predict q_z at arbitrary locations using measurable features shown in equation (1), especially when λ and $\frac{\partial T}{\partial z}$ are neither directly known nor continuous across the region. In such cases, we can train a ML-model to approximate:

$$q_z(\mathbf{x}, z) \approx \hat{f}(\mathbf{X})$$

where $\hat{f}(\cdot)$ is a data-driven mapping function learned from training data, and $\mathbf{X}$ is a feature vector built from multiple data sources. Let each observation (e.g., well or location) be indexed by i , then the feature vector $\mathbf{X}^{(i)} \in \mathbb{R}^p$ may include corrected temperature gradient, $g_T^i = \left(\frac{\partial T}{\partial z}\right)^i$, lithology fractions, ϕ_j^i (e.g., shale, sand, carbonate), over ground surface temperature, T_s^i . Additionally, such feature vector can include spatial location of the wells, (x^i, y^i) , depth-related features, z^i (e.g., cumulative thickness, or any derived properties), and estimated properties such as λ^i , and thermal resistance $R_T^{(i)}$. Thus, for each sample i , the feature space becomes, for example:

$$\mathbf{X}^i = [g_T^i, T_s^i, \phi_p^i, R^i, x^i, y^i, \lambda^i, z^i]$$

with the desired target variable, $y^i = q_z^i$ which can be either measured or calculated from Fourier law at the well location.

A Random Forest regressor is an ensemble learning method that builds multiple decision trees and averages their predictions to estimate a continuous target variable (Breiman 2001). With the target variable as vertical heat flux q_z , the model aims to learn a mapping: $\hat{f}: \mathbf{X} \rightarrow \hat{q}_z$ from the training dataset:

$\mathcal{D} = \{(\mathbf{X}^{(i)}, y^{(i)})\}_{i=1}^N$ where $\mathbf{X}^{(i)} \in \mathbb{R}^p$ is the feature vector for the i th observation, and N is the total number of

such training observations. Mathematically, a Random Forest Regressor learns a function (f) from training data $\{(\mathbf{X}^{(i)}, y^{(i)})\}_{i=1}^N$ such that:

$$\hat{f}(\mathbf{X}) = \frac{1}{M} \sum_{m=1}^M T_m(\mathbf{X}) \quad (2)$$

where M is the number of decision trees, and $T_m(\mathbf{X})$ is the output of the m^{th} tree trained on a bootstrap sample. After training all trees, the prediction for a new sample $\mathbf{X}^*$ is obtained by averaging the predictions of all trees (Breiman 2001):

$$\hat{q}_z^* = \hat{f}(\mathbf{X}^*) = \frac{1}{M} \sum_{m=1}^M T_m(\mathbf{X}^*) \quad (3)$$

Once trained, $\hat{f}(\mathbf{X})$ can predict heat flux q_z , at any new location. For regions with spatial gaps or heterogeneous geology, Random Forests can be constructed in conjunction with spatial interpolation methods (e.g., kriging on residuals) to generate continuous heat flow maps (Shahdi 2021). However, to account for spatial smoothness, ML-predicted results can be further post-processed using geostatistical interpolation techniques such as universal kriging or radial basis functions (RBF) as per the following formulation:

$$\hat{q}_z(\mathbf{x}, z) = \hat{f}(\mathbf{X}(\mathbf{x}, z)) + \epsilon_{\text{krig}}(\mathbf{x}, z) \quad (4)$$

Where $\hat{q}_z(\mathbf{x}, z)$ is the final predicted vertical heat flux at spatial location $\mathbf{x}=(x,y)$ and depth z , $\hat{f}(\mathbf{X}(\mathbf{x}, z))$ is the Random Forest predicted term and $\epsilon_{\text{krig}}(\mathbf{x}, z)$ is the geostatistical correction term, representing a residual adjustment obtained through kriging or any other interpolation technique. This additional term accounts for spatial correlation among prediction errors, and also for local biases in ML-predictions (Abdelmalek-Lee 2024). By leveraging both the RF regressor and kriging-like geospatial methodology, this combined approach encompasses the strengths of both methods: RF for capturing complex, non-linear relationships, while kriging ensures spatial correlation (Fahimuddin, 2025).

To measure the goodness of the Random Forest-trained model and ensure reliability of the predicted heat flux values at locations away from wells, several numerical metrics are used. The mean absolute error (MAE) computes the average absolute difference between the predicted and actual heat flux values across all training or test samples (Hastie 2009):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |q_z^{(i)} - \hat{q}_z^{(i)}|$$

Where $\hat{q}_z^{(i)} = \hat{f}(\mathbf{X}^{(i)})$ is the predicted heat flux using the Random Forest regressor, and $q_z^{(i)}$ is the actual (measured or calculated) heat flux at the i^{th} location. Similarly, the root mean squared error (RMSE) measures the square root of the average squared errors, penalizing larger deviations more strongly:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (q_z^{(i)} - \hat{q}_z^{(i)})^2}$$

Additionally, cross-validation assesses the model's generalization capability by partitioning the data into K folds (in this case, $K=5$). Here the model is trained on 4 folds and tested on the remaining fold. This process is repeated 5 times so that each fold serves once as the test set. The error metrics (e.g., MAE) are then averaged across all 5 folds to estimate the generalization performance (Hastie 2009). As a "reality check" to ensure ML-predictions align with the natural geospatial variability of the heat flux in the region, a numerical consistency checking with neighboring wells has also been investigated.

As proof of concept of the proposed methodology for regional heat flux prediction, a case study has been proposed in the form of a representative "synthetic use case". A key objective is to implement a data-driven methodology to predict the regional heat flux distribution by applying ML-methodology and associated geospatial interpolation, designed to simulate realistic subsurface conditions across multiple wells. The input data includes four distinct wells, each identified by a unique latitude and longitude coordinates. The geographical coordinates allow for spatial analysis and interpolation of heat flux values across the study area. For each well, a set of depth-temperature measurements is assigned by simulating borehole temperature profiles. Provided synthetic depth measurements span from 100 meters to 4200 meters across the wells, paired with corresponding temperature readings from 21°C to 197°C, which are critical for calculating geothermal gradients. Given surface temperatures vary between 25°C and 30°C to account for near-surface thermal conditions. Additional contextual features include surface temperature, rock type (e.g., shale, carbonate, sandstone, and crystalline basement), and formation age, which serve as both geological and thermal indicators. Rock types are further encoded into thermal conductivities using a predefined dictionary, which forms the basis for calculating effective thermal properties such as thermal resistance and thermal gradients. Additional derived features include depth-to-conductivity ratios and cumulative thermal resistance. These calculated "features", along with the raw inputs, are assembled into an extended feature vector, which outputs a design matrix that is used to train a Random Forest regression model. For consistency, corrected bottom-hole temperatures are obtained using Horner extrapolation to better estimate equilibrium conditions at well bore conditions (Fahimuddin 2024). Thus, the final feature space includes spatial coordinates, lithological encoding, temperature-depth ratios, surface temperature, thermal resistance, and derived parameters such as geothermal gradient and heat flow.

For the data mining and exploratory data analysis in this study, the RF regression algorithm from Scikit-Learn library has been used (Pedregosa 2011). The correlation matrix quantifies the linear relationships among raw and derived features such as depth, latitude, temperature, thermal conductivity, and thermal resistivity used in the case study. As seen from the left side of Figure 1, strong positive correlations are evident between depth and temperature, thermal resistance, and depth/ k , which can be explained by geophysical principles where deeper formations tend to be hotter and exhibit higher thermal resistance. Conversely, features like latitude and surface temperature display weak or even negative correlations with heat flow, suggesting limited linear dependence. However, the feature importance plot derived from the trained Random Forest model reveals a different perspective (right side of Figure 1). It highlights latitude and surface temperature as the dominant predictors of heat flow, with depth and other physically intuitive variables contributing much less. This discrepancy arises because Random Forest captures complex, nonlinear interactions and spatial dependencies that are not reflected in simple linear correlation coefficients as demonstrated by the correlation matrix.

As an example, latitude may act as a proxy for unmodeled geological factors such as regional geothermal gradients or tectonic setting, while surface temperature directly influences the computed gradient used in thermal modeling. This apparent contrast between statistical correlation and model-driven feature relevance underscores the importance of interpreting Random Forest outputs within the broader context of geospatial reasoning.

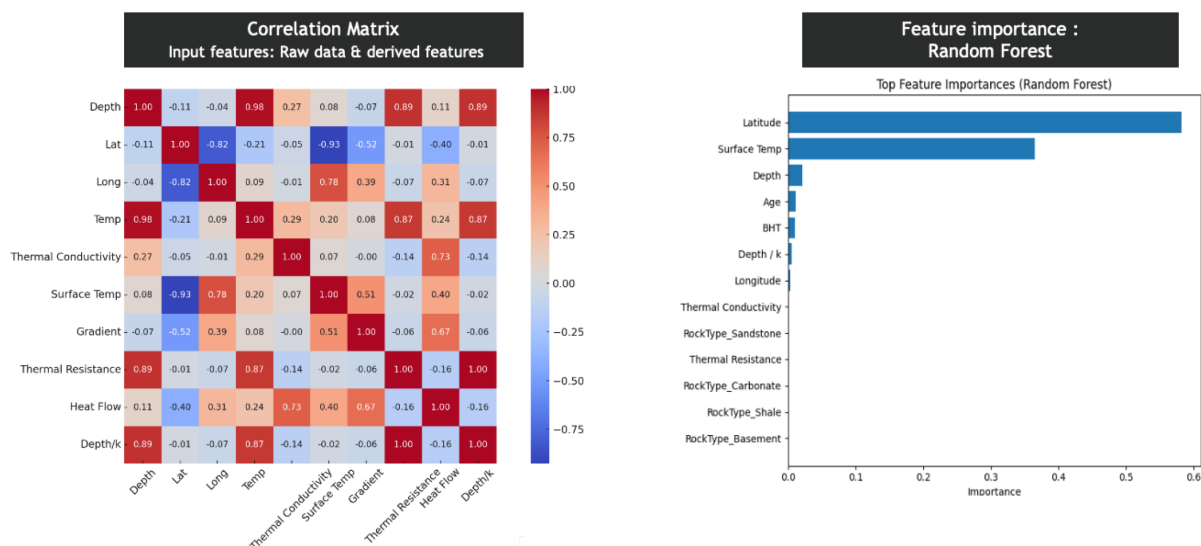


Figure 1—Correlation matrix and feature importance of Random Forest regressor used in the study.

The Random Forest (RF) model training and hyperparameter tuning has been used in this case by following a structured workflow that combines domain-specific feature engineering with cross-validation for optimal performance (Pedregosa et al., 2011). As the first step, a feature matrix ($\mathbf{X}$) has been constructed by extracting spatial information (latitude, longitude), depth, surface temperature, corrected bottom-hole temperatures (BHT), thermal conductivity, rock type encodings, and cumulative thermal resistance. For the target vector ($\mathbf{y}$), heat flow has been estimated from geothermal gradients and thermal conductivity. The synthetic dataset is then split into training and testing subsets using stratified sampling based on latitude bins to preserve spatial variability. ML-pipeline has been defined to standardize features using *StandardScaler* (mainly for compatibility and future model comparisons) and train a *RandomForestRegressor* as per the *Scikit learn* python library (REF). Hyperparameter tuning is conducted with *GridSearchCV*, performing a 3-fold cross-validation over a grid of optimized parameters to identify the best-performing model configuration. Both mean absolute error (MAE) and root Mean squared error (RMSE) metrics have been used to determine goodness of predicted heat flux. Also, a 5-fold cross-validation has been applied to report the "model generalization error" as per the formulation " $mean \pm standard deviation$ " of MAE scores. The spatial interpolation capability of the heat flux predictor has been ensured through a hybrid approach that integrates Radial Basis Function (RBF) interpolation with characteristics of kriging-like weighting, effectively capturing spatial variability and uncertainty. Specifically, an inverse distance weighting (IDW) function has been used where weights are computed as the inverse of distance raised to a power (typically 2), thus closely mimicking the nugget-structured covariance behavior in simple kriging. By blending RF model predictions with IDW results using weighted averaging (e.g., 70% RF and 30% IDW), the approach captures both feature-based learning and spatial coherence. The resulting hybrid surface is geologically consistent, honors well locations, and includes uncertainty visualization.

Results and Discussions

Figure 3 shows temperature-depth profiles for four wells used as training data for an RF model to predict regional heat flow. Each subplot corresponds to one well and presents a visual representation of the subsurface thermal and lithological structure, which are used as features in the RF training pipeline. The black curves with dots, represent Horner-corrected bottom-hole temperatures (BHT) using the equation:

$$T_{\text{corrected}} = T_{\text{measured}} + a \cdot \ln\left(\frac{t_c + t_s}{t_c}\right)$$

where t_c = circulation time and t_s = shut-in time (Deming 1989). The red dashed line represents a linear regression fit to the corrected temperatures relative to depth, accounting for surface temperature. The slope of this line is the geothermal gradient, and it is used to compute heat flow using, $q = \lambda \cdot \nabla T \cdot 1000$ (in mW/m²) where λ is the average thermal conductivity of rock layers in the well.

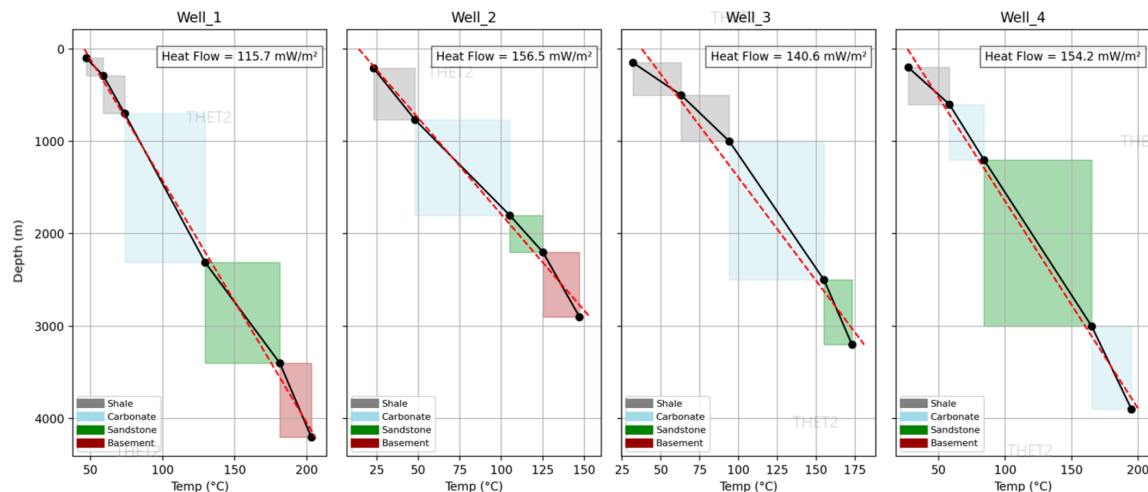


Figure 2—Temperature profile estimated at given 4 well locations (red dashed line: gradient; black solid line: Horner corrected available BHT data)

This heat flow value is displayed at the top of each subplot and is used as the target variable, y during RF training. Each lithological layer with color bands shows different rock types (shale, carbonate, sandstone and crystalline basement), and these are incorporated within the feature space vector X by encoding into "one-hot" categorical features. The regional temperature and heat flow trends inferred from the four training wells, as seen in Figure 3, reveal a distinct spatial variation in subsurface thermal conditions. A progressive increase in heat flow is observed from Well_1 toward Well_2 and Well_4, indicating a possible south-westward enhancement in thermal flux. Well-3 exhibits a relatively higher heat flow than Well-1, which is located to the southwest, but a lower heat flow than Well-2 and Well-4, which are situated to the west-northwest of Well-3 and Well-1. This suggests that the data actually confirm a higher thermal gradient towards the north in general, with a more pronounced increase towards the northwest than the northeast. This pattern implies that the thermal flux is not uniformly distributed, but rather exhibits a directional preference, with the northwest direction showing a more significant enhancement in heat flow. The presence of thermally conductive sandstone and carbonate units in the deeper sections of Wells 2 and 4 appear to enhance their heat flow signatures, while the uniformly shale-dominated upper layers across all wells serve as insulating units, which caused intensified temperature gradients at depth.

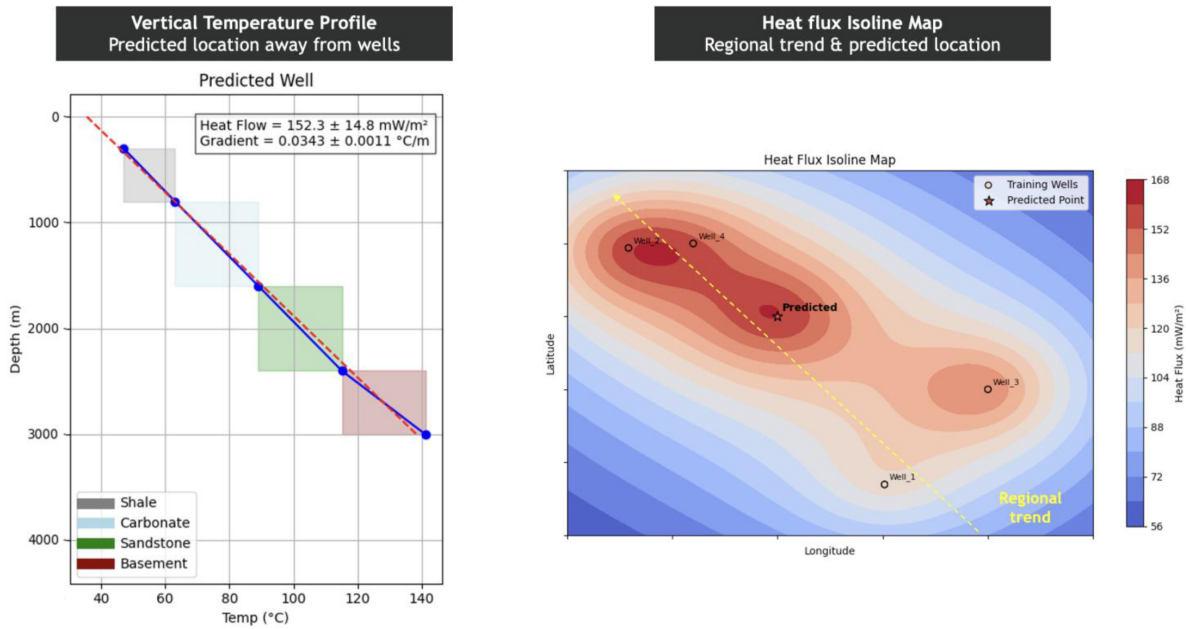


Figure 3—Temperature profile estimated at given 4 well locations (red dashed line: gradient; black solid line: Horner corrected available BHT data)

By capturing both linear and non-linear temperature trends, along with lithological heterogeneity, the trained model has the potential to gain predictive strength. As per the quality control (QC) of RF-trained model, the well-wise mean absolute error (MAE) stands at 0.89 mW/m² and root mean squared error (RMSE) of 1.15 mW/m². This suggests that the model captures the heat flow values with high accuracy relative to the physically-derived ground truth from geothermal gradients and rock thermal conductivities. The slightly higher RMSE compared to MAE implies that larger deviations are infrequent but present, likely due to local lithological variability or non-linear temperature profiles, as observed in Well_3. Additionally, the 5-fold cross-validation results yield a much lower average MAE of 0.38 ± 0.39 mW/m², demonstrating the model's strong generalizability and robustness. The low standard deviation in cross-validation MAE further confirms that the model performs consistently across different training subsets, minimizing overfitting and ensuring reliable heat flow predictions when extrapolated to new spatial locations. To enhance spatial realism and prevent model overfitting, the RF prediction (q_{rf}) is blended with the spatially interpolated inverse distance weighting (q_{idw}) estimate using weights, thus the equation (4) becomes:

$$\hat{q}_z(\mathbf{x}, z) = w_{RF} \cdot \mathbf{q}_{rf} + w_{IDW} \cdot \mathbf{q}_{idw}$$

where $w_{RF} = 0.7$ and $w_{IDW} = 0.3$. This hybrid model leverages both local spatial trends and multi-feature learning, with higher weights being placed in the RF regressor. This dual formulation ensures that the new well prediction remains consistent with the thermal-lithological trends learned from the training data, while honoring the spatial structure revealed by known well locations. Additionally, the radial basis function (RBF) interpolator was used to construct a continuous spatial heat flow map from discrete well predictions, including both the training wells and the newly predicted well. The temperature-depth profile of the predicted well (left side of Figure 3) aligns well with the established regional thermal trend observed across the four training wells (see Figure 3). The calculated gradient of 0.0343 ± 0.0011 °C/m is consistent with those derived from high-heat-flow training wells, particularly well_2 and well_4, suggesting a similarly conductive thermal regime. The estimated heat flow of 152.3 ± 14.8 mW/m² also falls within the upper range observed in the training dataset, reinforcing the notion that this predicted location lies within a regional high-heat-flow corridor. The predicted well intersects four main lithologies in sequence, mirroring the

vertical stratigraphy observed in several training wells. These rock types, especially the deeper sandstone and basement units, play a key role in shaping the thermal conductivity profile used in calculating the predicted heat flow. The regional heat flux isoline map (see right side of Figure 3) provides a coherent spatial framework to contextualize the predicted well's heat flow estimate. The predicted well lies within a localized high-heat-flux corridor defined by an interpolated zone of 150–160 mW/m², as delineated by the regional trend "arrow". The predicted position is consistent with the heat flow estimate of 152.3 mW/m², derived from the well's corrected temperature profile and Random Forest regression. This demonstrates a strong alignment between point-wise prediction and regional interpolation.

Conclusion

The proposed heat flow prediction methodology addresses a potential data-driven approach for mineral exploration from geothermal brines by identifying high heat flow zones and by characterizing subsurface thermal and lithological conditions. These capabilities enable geoscientists to target regions with thermal energy and geological settings that are conducive to brine-hosted mineral deposits, especially in frontier exploration areas where there is limited temperature data available. The aim of this paper is to demonstrate an ML-powered workflow to predict a regional heat flux through a synthetic dataset with a limited number of wells. Due to the lack of a larger and practical dataset, it is evident that the RF algorithm has the potential to overfit the predicted temperature profile at new locations. In this case, IDW assumes that spatial proximity implies similarity, which breaks down in geologically heterogeneous settings where nearby wells can have drastically different thermal regimes due to faults or lithological changes. Also fixed weighting ($w_{RF}=0.7$ and $w_{IDW}=0.3$) assumes a uniform reliability of RF and IDW across the region. However, optimal weights are likely spatially variable, especially in data-sparse or geologically complex zones. It is assumed here that thermal conductivity is fixed per rock type used, ignoring intra-lithology variability and temperature/pressure dependence. In spite of several limitations, the proposed workflow is simple to implement using open-source python-based software components. Hence, this methodology could be regarded as a stepping stone in establishing a data-driven techniques to predict regional heat flow regimes at scale.

References

- Abdelmalek-Lee, E., Burton, H. 2024. "A dual Kriging-XGBoost model for reconstructing building seismic responses using strong motion data." *Bulletin of Earthquake Engineering* 3563–3589.
- Aljubran, M., Horne, R. 2024. "Thermal Earth model for the conterminous United States using an interpolative physics-informed graph neural network." *Geothermal Energy*.
- Breiman, L. 2001. "Random forests." *Machine Learning* <https://doi.org/10.1023/A:1010933404324>, 45(1):5–32.
- Cetin, R., Azdin, H., Haklidir, T. 2024. "Mineral Extraction from Geothermal Reservoirs: A Case Study from Western Anatolia." PROCEEDINGS, 49 th Workshop on Geothermal Reservoir Engineering. Stanford, California: Stanford University.
- Deming, D. 1989. "Application of bottom-hole temperature corrections in geothermal studies." *Geothermics*. 18(5-6):775–86.
- Dobson, P., Gasperikova, E., Borgin, S. 2023. "U.S. DOE Clean Energy Demonstration Program on Current and Former Mine Land - A Review of Geothermal Energy Case Studies and Opportunities." PROCEEDINGS, 48 th Workshop on Geothermal Reservoir Engineering. Stanford, California: Stanford University.
- Engle, M., Nze, C., Neupane, G. 2022. "Predicting Rare Earth Element Potential in Produced and Geothermal Waters of the United States via Emergent Self-Organizing Maps." *Energies* 15(13).
- Fahimuddin, A., Birkle, P., Saeed, W. 2025. "Geospatial prediction of geochemical properties of produced waters amples: Application of dual kriging - gradient tree boosting method." MEOS GEO 2025 - Middle East Oil, Gas and Geosciences Show. Bahrain: Society of Petroleum Engineers (SPE). Accepted for publication.
- Fahimuddin, A., Hussein, W., Anifowose, F. 2024. "Data Driven Subsurface Workflow for Geothermal Resource Exploration: Applications of Machine Learning Methodology." Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC 2024). Abu Dhabi.
- Hastie, T., Tibshirani, R., Friedman, J. 2009. *The Elements of Statistical Learning Data Mining, Inference, and Prediction*, Second Edition. New York: Springer.

- Mudunuru, M., Vesselinov, V., Ahmmed, B. 2022. "GeoThermalCloud: Machine Learning for Geothermal Resource Exploration." *Journal of Machine Learning for Modeling and Computing* **3**(4).
- Okoroafor, R. E., Smith, C. M., Ochie, K. I., Chinedu, J. N., Gudmundsdottir, H., & Aljubran, M. 2022. "Machine Learning in Subsurface Geothermal Energy: Two Decades in Review." *Geothermics*.
- Palumbo, F., Main, I. 1999. "The thermal evolution of sedimentary basins and its effect on the maturation of hydrocarbons." *Geophysical Journal International*, Volume **139**, 248–260.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Perrot, M. 2011. "Scikit-learn: Machine Learning in Python." *The Journal of Machine Learning Research*.
- Qiu, N., Chang, J., Zhu, C. 2022. "Thermal regime of sedimentary basins in the Tarim, Upper Yangtze and North China Cratons, China." *Earth-Science Reviews*, Volume **224**,.
- Shahdi, A., Lee, S., Karpatne, A., Nojabaei, B.. 2021. "Exploratory analysis of machine learning methods in predicting subsurface temperature and geothermal gradient of Northeastern United States." *Geotherm. Energy* **9** (18) <https://doi.org/10.1186/s40517-02>.
- Sircar, A., Yadav, K., Oza, H. 2021. "Application of machine learning and artificial intelligence in oil and gas industry." *Petroleum Research* 379–391.
- Smith, Y., Kumar, P., Mcleannan, J. 2017. "On the Extraction of Rare Earth Elements from Geothermal Brines." *Resources*, Volume **6**,.
- Svetnik, V., Liaw, A., Tong, C. Feuston, B. 2003. "Random forest: a classification and regression tool for compound classification and QSAR modeling." *J Chem Inf Comput Sci*; **43**(6):1947–58. doi: [10.1021/ci034160g](https://doi.org/10.1021/ci034160g).
- Szanyl, J., Rzbach, L., Abdulhaq, H. 2023. "Geothermal Energy and Its Potential for Critical Metal Extraction-A Review." *Energies* **16**(20), 7168.
- Zhang, H., Li, K., Shi, S. 2025. "Prediction of Geothermal Heat Flow Using XGBoost with the Applications in Coupling Coalbed Methane and Geothermal Resource." *ACS Omega*.